

Past, Current and Future Trend in System Software Methodology

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Abstract— This paper examines the evolution of system software development methodologies from traditional to modern approaches, analyzing the strengths and limitations of each in adapting to technological and market demands. Early methodologies, including the Waterfall and Spiral models, provided structured frameworks focused on sequential processes but often lacking flexibility. As technology advanced, Agile and DevOps methodologies emerged, emphasizing iterative development, continuous delivery, and enhanced collaboration between development and operations teams. These modern methodologies have proven effective for dynamic project environments and fast-changing customer needs. Looking toward the future, emerging methodologies driven by Artificial Intelligence (AI) and Quantum Computing promise to enhance adaptability, risk prediction, and efficiency through data-driven automation and advanced computational capabilities. This paper offers a comprehensive overview of these developments, highlighting the lacking from past methods that are perfected by the modern methods and how future methods can enhance system software development to even higher levels.

Keywords— Spiral Model, Risk Management, DevOps, Quantum Computing, Waterfall Model, Agile Methodology, AI-Assisted Dev, Artificial Intelligence

I. INTRODUCTION

Due to the rapid development of system software, engineers need a strategic plan to develop a successful final system with a complete documentation of each phase and suitable to be used in a team environment where collaboration is inevitable which causes the birth to a variety of system software methodology over the years. System software development methodologies have evolved to meet the growing demands of technology, shifting from rigid, sequential models to more adaptive and iterative approaches. Traditional models like Waterfall and Spiral laid the groundwork, providing structured, predictable processes but often lacking flexibility. With modern needs for rapid delivery and client responsiveness, Agile and DevOps emerged, emphasizing continuous integration and customer satisfaction. Looking ahead, future methodologies are promising even greater efficiency and convenience due to the rise of Artificial Intelligence and Quantum Computing.

II. PAST TREND IN SYSTEM SOFTWARE METHODOLOGY

This section will explain the traditional methodology used in developing system software in the past. These methods were designed to ensure a project conducted in sequential processes, easy to monitor any progress and able to produce a successful final product.

A. Waterfall Model

The Waterfall Model was the first Process Model that was introduced in software development strategy [1][3] credited to Winston W. Royce in 1970 [2]. It is a simple and easy to use methodology that includes five distinct phases which are Analysis, Design, Implementation, Testing, and Maintenance[1]. In the Waterfall model, each step must be fully completed before the next step can begin which causes no overlapping between each phase[1].

This model is best used when all the requirements are laid out clearly and understood perfectly by all parties [2] without any further addition to the agreed requirements. The Waterfall model assumes that once the initial requirements are set and all goals are clarified, the development team has a clear path to complete the project [3]. However, in most real-life cases, this is not true, as customers may change their preferences for certain features, requiring some or all phases to be re-evaluated. This leads to additional costs and time spent on various parts of the project, which could, in turn, lower customer satisfaction.

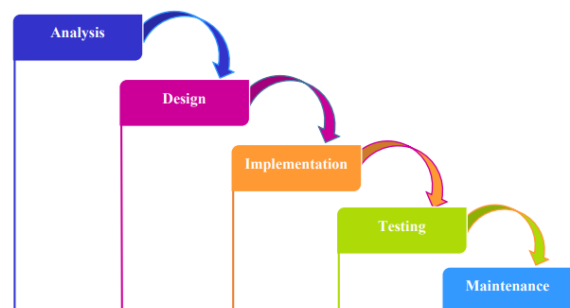


Figure 1: Waterfall Model

B. Spiral Model

According to [11] the spiral model was first introduced in the 1980s by Boehm where the process is represented in spiral with each loop split into four sectors which is objective setting, risk assessment and reduction, development and validation, and planning. The spiral model combines an iterative development with structured aspects of the waterfall model, but the difference between the two models is its explicit recognition of risk [11]. The spiral model may be adaptable and allows flexibility in system software development. However, this methodology is not widely used. This methodology was designed for high-risk and large projects in mind and not suitable for projects with simple requirements and limited budget. Moreover, an experienced and skilled project manager is needed in managing the risk assessment in order to maintain the functionality of the system.

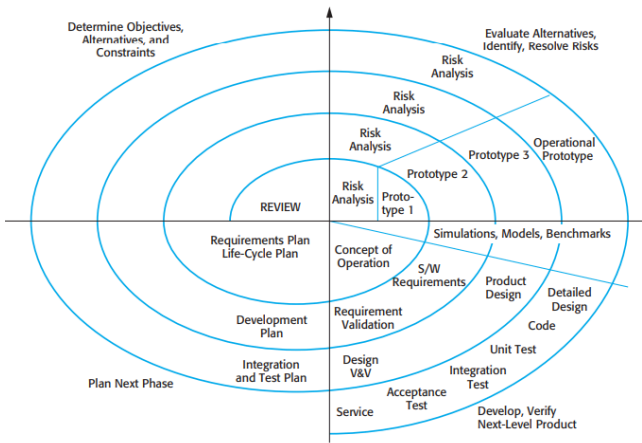


Figure 2: Spiral Model

III. CURRENT TREND IN SYSTEM SOFTWARE METHODOLOGY

As technology evolves, increasingly complex challenges and feature demands arise in software development. This has led engineers to create new development models that better align with modern client needs and optimize company budget and time.

A. Agile Methodology

In February 2001, 17 software engineering consultants had discussed commonalities between their respective methodology which then classified as agile [2]. Agile methods vary, but they all share core values and principles outlined in the Agile Manifesto. This manifesto emphasizes twelve principles, some of them are satisfying the customer and welcoming change [2]. There are many different models of Agile, some of them are Extreme Programming (XP) and Scrum [4].

Agile models are designed for effective response to change, aiming to achieve rapid, periodic software delivery through incremental and iterative development [2]. This approach repeats phases until the customer is satisfied with the final product, using multiple smaller processes, or "iterations," that incorporate all development phases. Agile methods, a subset of iterative and evolutionary processes [4], rely on iterative enhancement and opportunistic development.

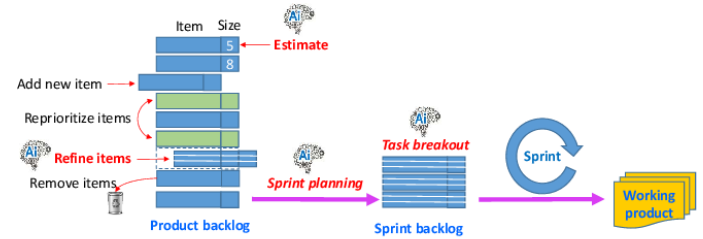


Figure 3: A typical Agile Process

B. DevOps

Patrick Debois was the first person to coin the term "DevOps" after his talk at the Agile Toronto 2008 conference where he presented his IEEE paper [9]. After some time, DevOps methodology has been evolving from the principles of agile where it combines development (Dev) and operations (Ops) making it a tool that is able to deliver a software system within a short amount of time with high quality [10]. As startups often practice DevOps due to its flexibility and speed, DevOps is also being used widely across organizations, especially well-established organizations.

IV. FUTURE TREND IN SYSTEM SOFTWARE METHODOLOGY

Future trends in software development methodology are increasingly driven by the need for speed, adaptability, and quality in a rapidly evolving tech landscape. As software continues to shape industries, future methodologies will likely prioritize collaboration, user-centered design, and continuous improvement to meet the demands of both developers and users.

A. Artificial Intelligence Assisted Dev

Artificial Intelligence (AI) is making a substantial impact on various sectors and is expected to disrupt production, manufacturing, and delivery processes [5]. The rise of AI is fueled by the availability of big data, advances in AI algorithms like deep learning, and significant increases in computational power. The widespread use of software has generated vast amounts of data on software projects, which AI can leverage. We anticipate that AI will transform project management by automating basic administrative tasks, providing analytics-driven risk predictions, aiding in project planning, and offering actionable recommendations. The

current challenges and lack of effective tools present an opportunity for AI to enhance project management practices like Agile[5]. AI-driven tools can process extensive data from software projects, extract valuable insights, and learn to perform complex tasks such as effort estimation, task refinement, resource management, and sprint planning.

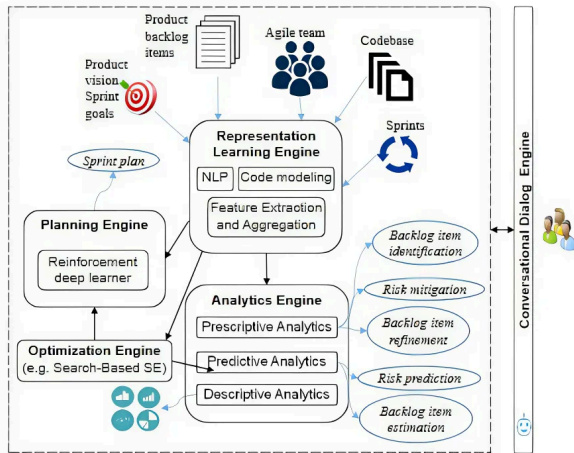


Figure 4: The architecture of AI-powered agile project management assistant

B. Quantum Computing

According to [6] Richard Feynman was the first person to consider applying the quantum mechanical principles to operate on quantum computers, where it also states that the case quantum mechanic cannot be simulated efficiently using classical computers. This can be proven based on a concept introduced by [6] where quantum computers are able to solve certain problems more efficiently than classical computers. Quantum computing has been getting its recognition as it introduced new concepts of traditional programming methods as it applied a unique behavior of quantum physics into computing [9]. As quantum computing keeps on evolving, it has been developed with a variety of technologies which the result will bring a very big impact to daily lives. According to [7] there are various dimensions of quantum computing and the examples for dimensions were explained through Figure 5.

Dimension	Examples
Applications	Radiotherapy optimization, speeding up AI algorithms, empowering modeling and simulations in aerospace and physics, drug discovery, vaccine development, cryptography, portfolio management, among others.
QC programming languages	OpenQL by TU Delft Netherlands, Silq by ETH Zürich Switzerland, Q# by Microsoft, Qiskit by IBM, Cirq by Google
QC platforms	Quantum Inspire from QuTech Netherlands, Microsoft Quantum computing platform, and IBM Quantum Experience
Open source software	Quantum compilers: BQSKit, D-Wave's qbsolv Computer simulators: QuEST, QuPy Editors: ProjectQ, Qiskit Circuit Composer
Industry	Norwegian Quantum Computing Group, IQM, Cambridge Computing, Qbce.eu, Aegiq, Algorithmiq, QBaltic, Arqit
Events	Quantum Software Engineering Workshops co-located with ICSE, QC Talks at University of Porto, Portugal; Quantum Software Engineering and Technology Workshop; International Workshop on the QuANtum SoftWare Engineering & pRogramming

Figure 5: Dimensions of quantum computing with example

V. CONCLUSIONS

System software methodology has been evolving for over the century based on the demands from over different organisations. From the evolution of the methodology, traditional methodology such as Waterfall and Spiral Model has been set as the most used methodology. However, people often struggle to adapt to it due to its risk.

This is being said with the emerging technology like AI and Quantum Computing that have potential in changing human daily lives. Hence, as the future system software methodologies continue to evolve, it will definitely lead to more adaptable system software development practices in order to rapidly produce more advanced tech that would keep up with rapid changing of the “Tech-World”.

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